

physical objects into blockchain services by building blockchain-specific hardware compatible with any IoT device. It helps to define permission for reading and writing rights into the IoT device. BigchainDB can be integrated with every IoT scenario where there is a need for immutability and tamper-proof storage of data assets, along with search and query capability with high throughput.

MySQL: This open source relational database management system brings data consistency, scalability, high performance, availability, and flexibility to IoT solutions by efficient collection of data from IoT devices. It helps in data transformation through annotation and aggregation making it easier to understand data better.

Based on the IoT solution requirement, if the data format is fixed then MySQL is the preferred database.

GridDB: This is a container data model extending the NoSQL key-value store, and representing data in the form of a collection referenced by keys. It provides high scalability, reliability, and availability.

The two types of containers in GridDB are:

- *Collection container:* General-purpose container
- *TimeSeries container:* Data associated with timestamp providing functions like data compression and data aggregation

Redis: This is an in-memory open source database. It is a popular choice for IoT solutions as a hot database. It is widely used by IoT solutions for data ingest real-time analytics, messaging, caching, and many other use cases. It helps in edge computing involving deep learning, image recognition and other innovative computing requirements.

CrateDB: CrateDB is an open source distributed SQL database management system that fully integrates a searchable document-oriented data store. The CrateDB

platform provides the distributed SQL query engine for faster joins, aggregations, and *ad-hoc* queries. This highly scalable and available database supports various types of data.

Cassandra: This is a highly scalable and distributed open source database for managing voluminous amounts of structured data across many commodity servers. It provides availability, linear scale performance, simplicity and easy distribution of data across multiple database servers. It supports strong data consistency across distributed architecture.

Hadoop: This open source software platform for distributed storage and distributed processing of very large data sets on computer clusters is built from commodity hardware. It helps in driving analytics from all IoT data. It easily ingests data from multiple data sources and supports both batch as well as real-time data ingest from sensors using tools such as Apache Kafka and Apache Flume. It handles multiple IoT data types, structures, and schemas. It supports real-time processing and applications on streaming data. Hadoop is a flexible, scalable and secure database.

A number of leading organisations, including leading automotive manufacturers, utilities, industrial automation companies, insurers, healthcare organisations, and telecom and technology leaders are adopting Hadoop.

Benefits of using open source databases in IoT

Open source databases in IoT have the following advantages over proprietary databases:

- Easy upgrade to new technologies with open source DB
- Ability to connect with upcoming device protocols and backend applications.
- Lower overall software cost, ease of change in technology, and open source APIs for integration
- Flexibility and easy change in architecture of solutions that are centred on microservices
- Flexibility to change with a change in the cloud service provider

IoT is one of the most significant sources of Big Data. This data is rendered useless without analytics power. It is very important to choose the right set of open source databases for an IoT solution, as there are so many available in the market.

We need to first analyse the business problem, arrive at a solution, break this solution into services, and understand the database needs of these services. This will help to narrow down the database choices.

Most IoT solutions can depend on a hot database for real-time data collection, processing, messaging and analytics. Cold databases are better suited to store historical data and gather business intelligence. This will make the architecture simple, lean and robust. **END** 🐧

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